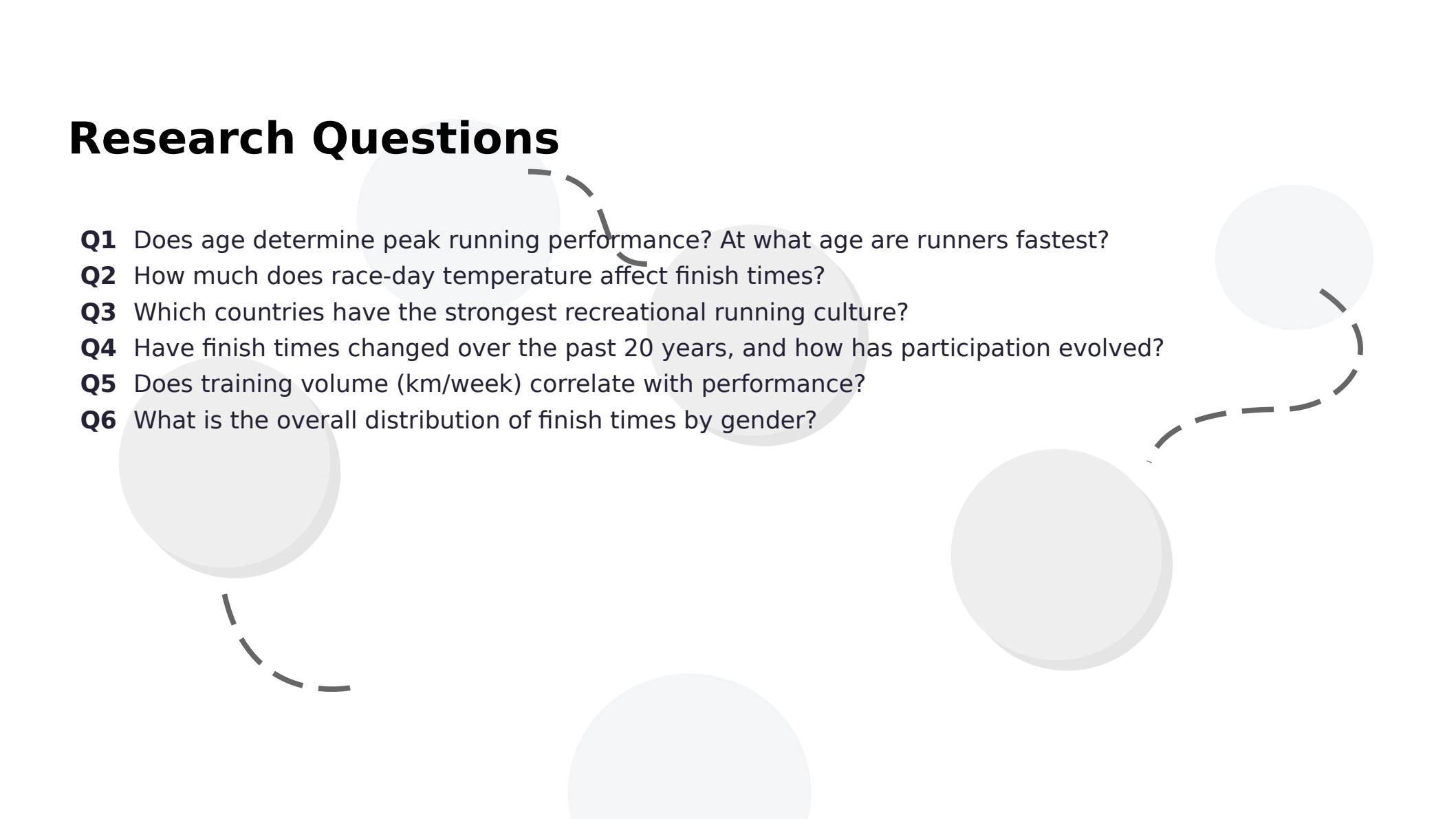


# **What Makes a Runner Fast?**

Ulysse Auplat Vijay Singh  
Università degli Studi di Milano 2025-2026

# Research Questions



- Q1** Does age determine peak running performance? At what age are runners fastest?
- Q2** How much does race-day temperature affect finish times?
- Q3** Which countries have the strongest recreational running culture?
- Q4** Have finish times changed over the past 20 years, and how has participation evolved?
- Q5** Does training volume (km/week) correlate with performance?
- Q6** What is the overall distribution of finish times by gender?

# Three Datasets · One Unified View

**01 Marathon Results** 2000-2019 Zenodo CSV 353 MB  
2.85 million finishers 10 major US races

Key columns: finish\_time, age, gender, city, year

**02 Strava Training Data** 36,412 athletes Kaggle CSV

Runners targeting major marathons worldwide

Key columns: distance, country, gender, age\_group, major

**03 Race-Day Weather** Open-Meteo API JSON free, no key

Historical daily data per city and race date

Key columns: temp\_mean\_c, precip\_mm, wind\_kmh

# Preprocessing

**Finish times** converted from HH:MM:SS strings to minutes; extreme outliers removed (<120 min or >600 min).

**Age groups** derived from raw age into four buckets (<30, 30-49, 50-69, 70+) for the Strava join.

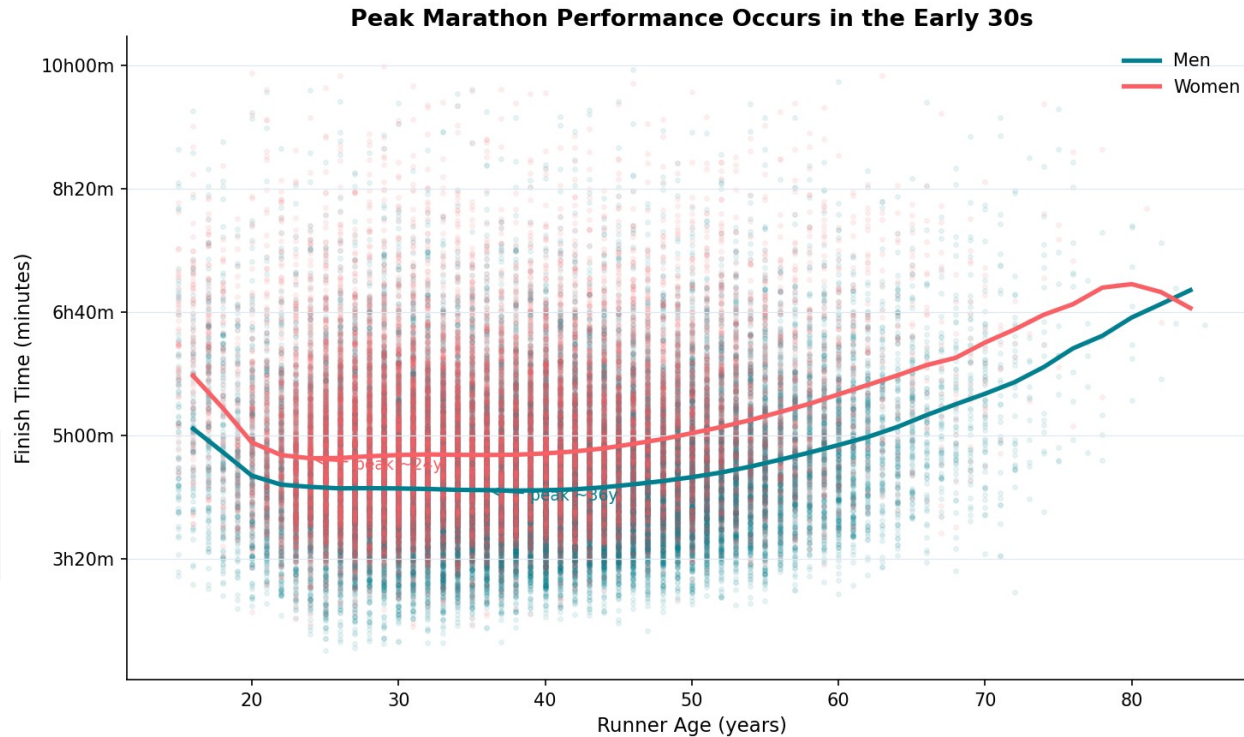
**City names** standardised across the two datasets so the join keys match correctly.

**Weather** fetched from Open-Meteo API for each (city, race date) pair and cached locally.

**Strava join** no shared athlete ID exists, so training data was merged at group level: for each gender × age group × race × year bucket, we took the median weekly km of matching Strava athletes and assigned it to every finisher in that bucket.

**Output** master.csv 2.85 M rows, 25 columns, ready for analysis.

## Q1 · Peak marathon performance occurs in the early 30s

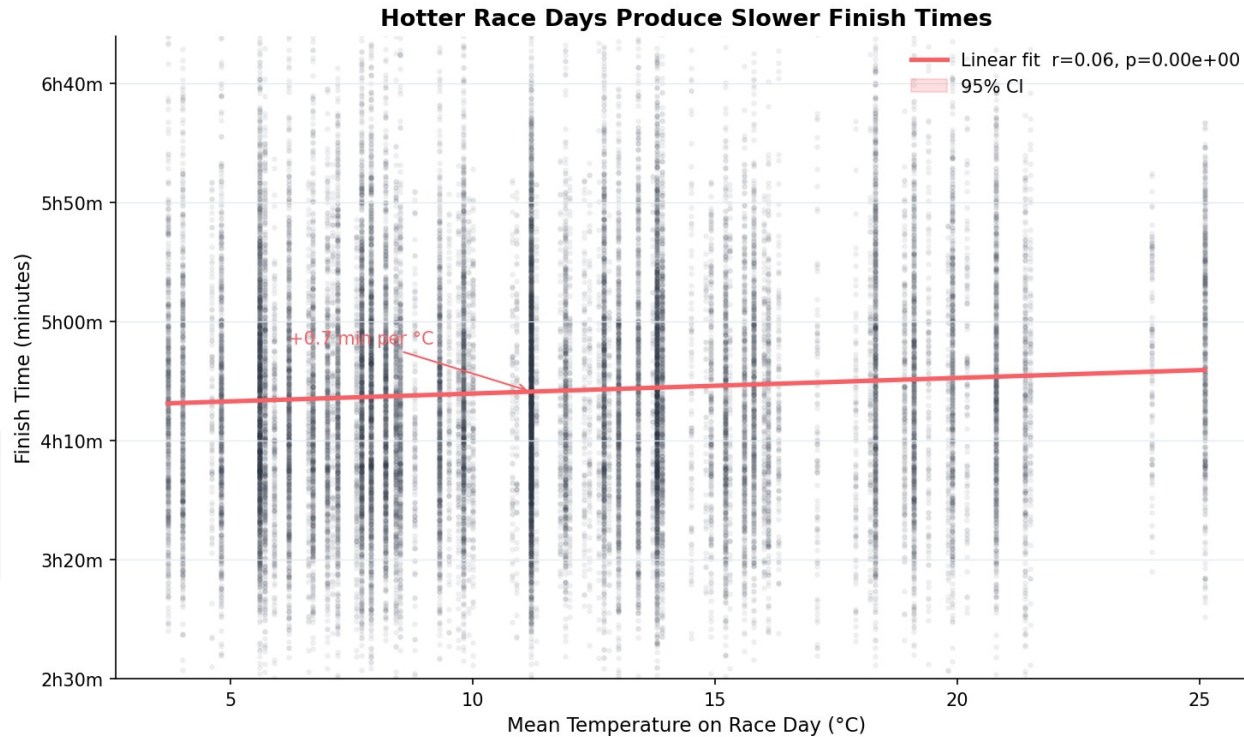


### Key Finding

Both men and women peak between ages 28–34.

After 35 the decline accelerates by age 70, runners take over an hour longer than at their peak.

## Q2 · Hotter race days produce slower finish times



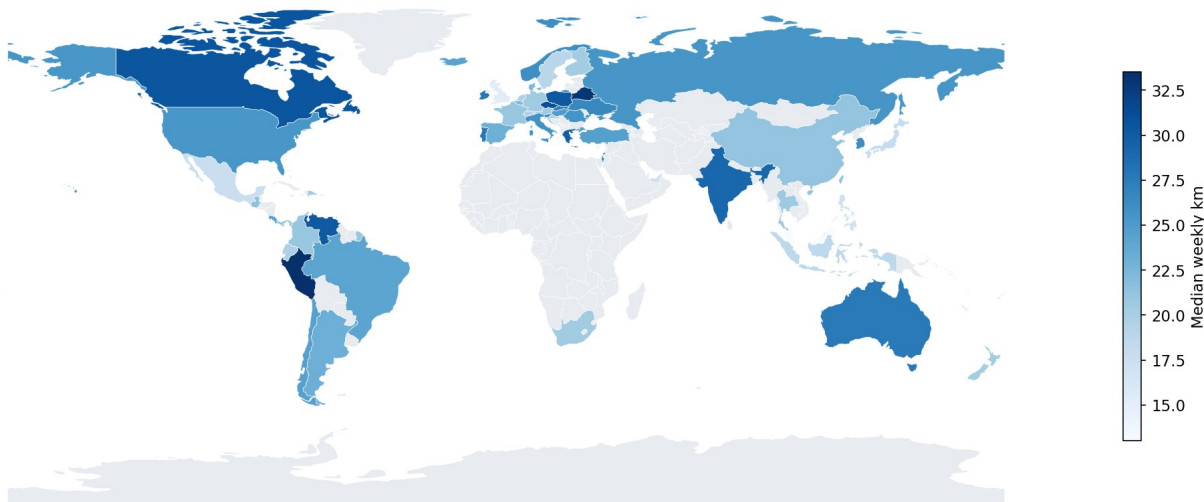
### Key Finding

Race-day temperature correlates with finish time at  $r = 0.063$ .

There is an effect but its not much.

### Q3 · Running culture by country — median weekly training volume

Running Culture by Country — Median Weekly Training Volume  
(Strava 2019 · marathon-targeting athletes · ≥20 unique athletes per country)



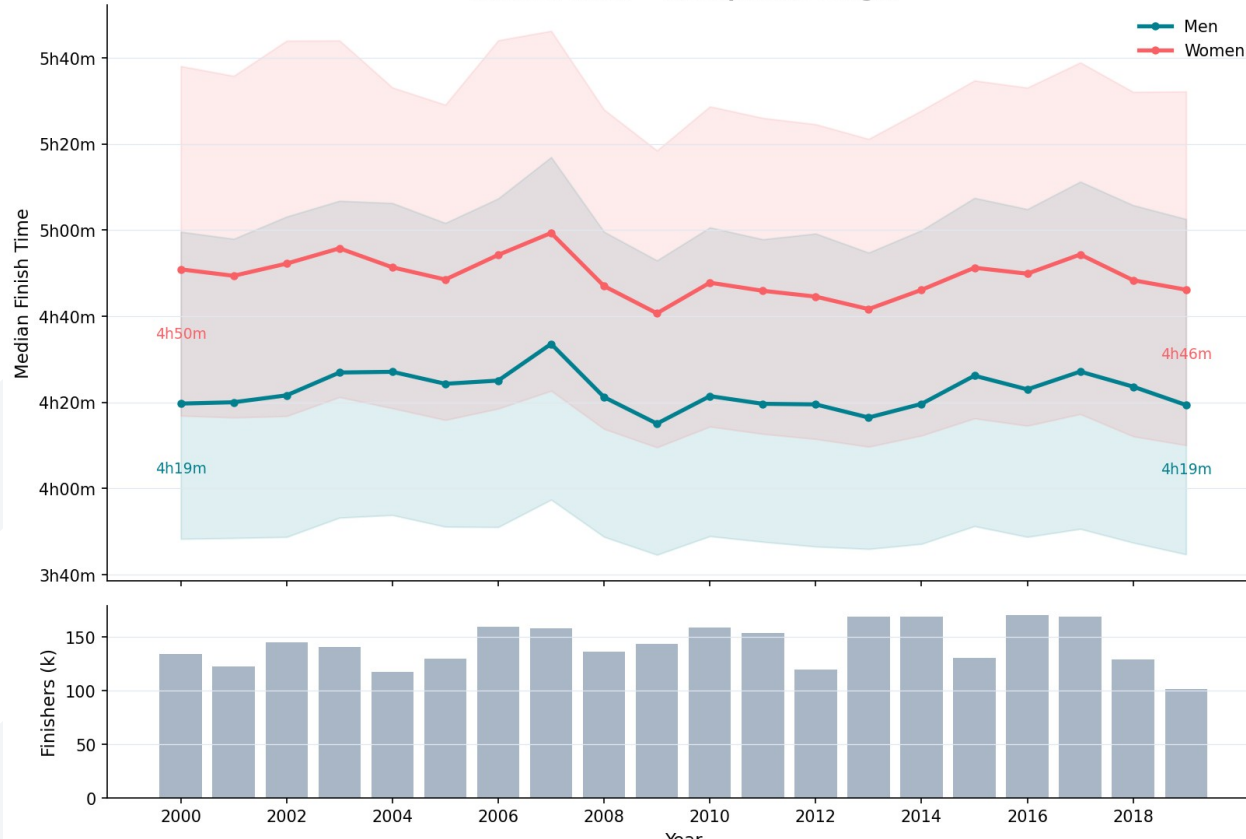
#### Key Finding

Peru, Canada, and Central Europe lead in median weekly training km among Strava runners targeting a major marathon.

Countries with fewer than 20 unique athletes are shown in grey.

## Q4 • Finish times have stayed flat over 20 years as participation fluctuated

Median Finish Times Stable Over 20 Years  
(shaded band = interquartile range)

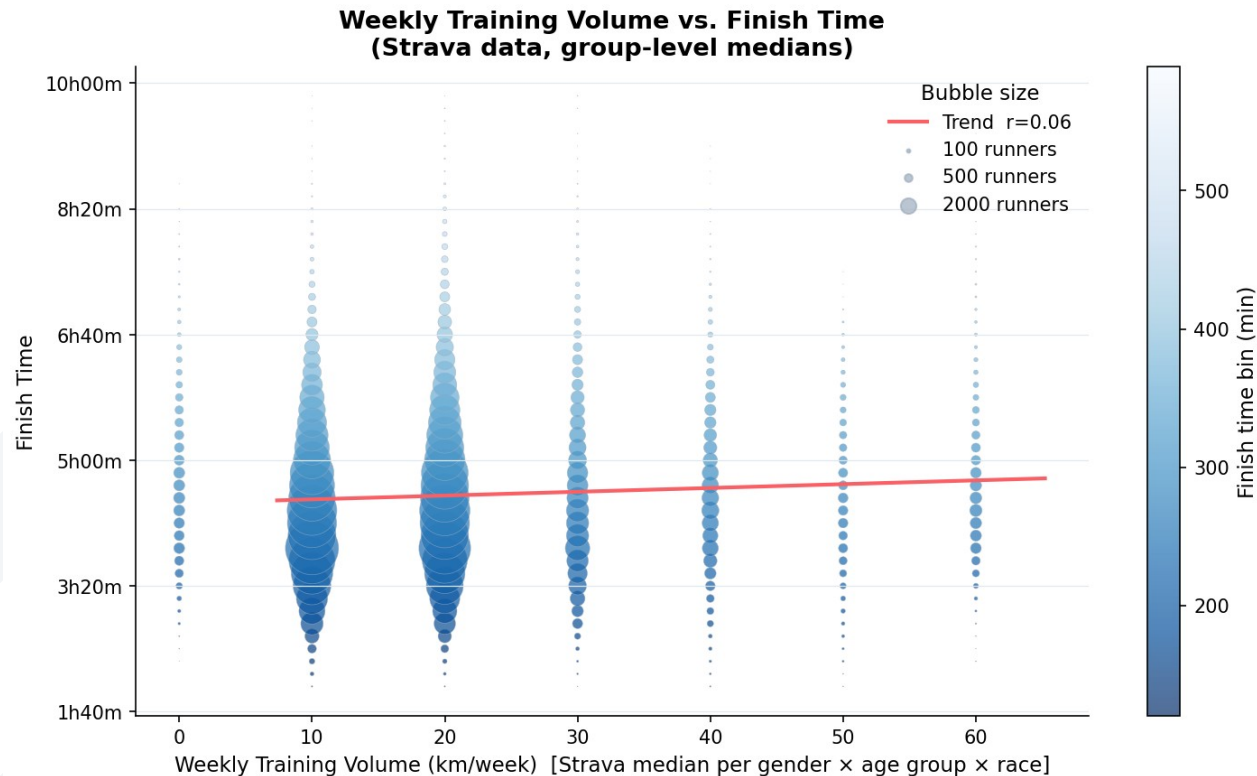


### Key Finding

Finish times have almost not changed  
The number of finishers fluctuates but  
we cannot see any trend.



## Q5 · Training volume (km/week) shows only a weak correlation with performance

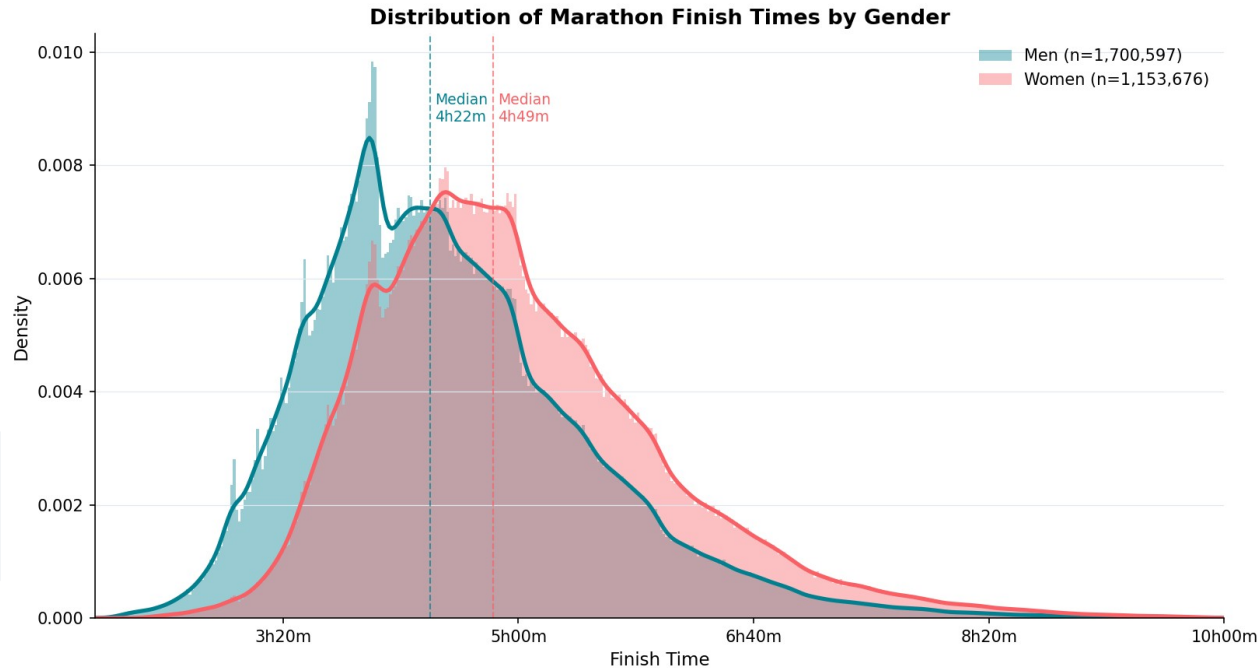


### Key Finding

$r = 0.06$  not much

Big variance at every training level suggests raw volume is not a good predictor.

## Q6 · Finish time distributions: right-skewed, women wider



### Key Finding

Men median: 4h22m (n = 1.7 M)

Women median: 4h49m (n = 1.15 M)

Both distributions are right-skewed.  
Women's curve is wider across all ability levels.

# What Makes a Runner Fast? Key Takeaways

**Age:** Peak performance in the early 30s then decline.

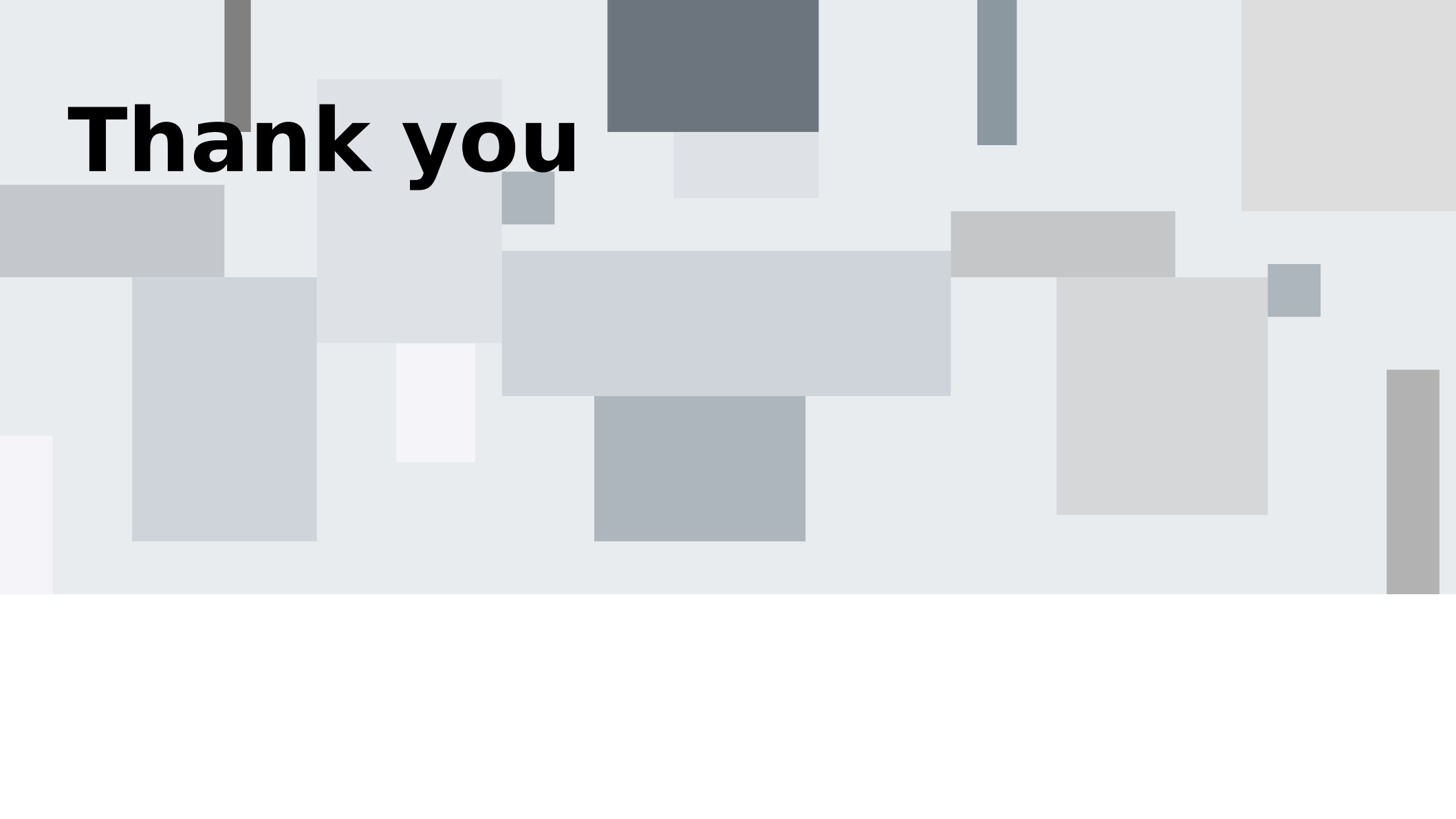
**Weather:** Race-day temperature plays a role but is not that important.

**Culture:** Peru, Canada, and Central Europe log the highest training volumes in Strava data.

**Time:** median finish time haven't changed.

**Volume:** More km/week doesn't imply much on the finish times.

**Gender gap:** Men 4h22m vs women 4h49m at the median.

The background of the slide is composed of numerous overlapping rectangular blocks in various shades of gray, from light to dark, and some white. These blocks are scattered across the upper half of the image, creating a modern, architectural feel. The text 'Thank you' is prominently displayed in the upper left area, rendered in a bold, black, sans-serif typeface.

**Thank you**